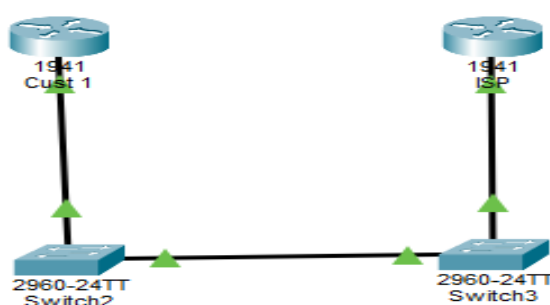


TP 3 : Configuration of a PPPoE Client Router for DSL Connection

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Objective

The objective of this lab is to simulate a WAN connection between a client and an Internet Service Provider (ISP) using the PPPoE protocol. This lab helps to understand PPP authentication and dynamic IP address assignment in DSL connections.

2. Network Topology

The network topology used in this lab consists of:

- Two routers (ISP and Cust1)
- Two switches
- Ethernet cables connecting the devices

The ISP router represents the Internet Service Provider, while the Cust1 router acts as the client connecting to the ISP through PPPoE.

ISP Router Configuration :

1. Creation of the PPPoE user

```
ISP(config)# username Cust1 password ciscoppoe
```

Why :

Cust1 = client name (must be the same as the one used on the client router).

ciscoppoe = CHAP password required in the lab.

Used for **PPP CHAP authentication**.

Creation of the IP address pool:

```
ISP(config)# ip local pool PPPoE_POOL 10.0.0.1 10.0.0.10
```

Why :

PPPoE_POOL = name of the pool.

10.0.0.1 → 10.0.0.10 = IP range for 10 clients as required in the lab.

These IP addresses will be **dynamically assigned by PPP**.

Creation of Virtual Template

- ISP(config)# interface virtual-template 1
ISP(config-if)# ip unnumbered GigabitEthernet0/1
ISP(config-if)# mtu 1492
ISP(config-if)# peer default ip address pool PPPoE_POOL
ISP(config-if)# ppp authentication chap
ISP(config-if)# exit

! Le Virtual Template utilise l'IP de **G0/1**.

! PPPoE réduit la MTU de **1500 → 1492**.

! indique quel **pool d'IP donner aux clients**.

! active authentication CHAP.

Creation of the PPPoE group :

- ISP(config)# bba-group pppoe GLOBAL
ISP(config-bba-group)# virtual-template 1
ISP(config-bba-group)# exit

!GLOBAL = nom du groupe PPPoE.

!On associe Virtual Template 1 à ce groupe.

Enabling PPPoE on the interface :

```
ISP(config)# interface Gig0/1
```

```
ISP(config-if)# pppoe enable group GLOBAL      ! active PPPoE sur cette interface.
```

```
ISP(config-if)# no shutdown
```

Cust1 Router Configuration :

Configuration of G0/1 :

- Cust1(config)# interface Gig0/1
Cust1(config-if)# pppoe enable
Cust1(config-if)# pppoe-client dial-pool-number 1
Cust1(config-if)# exit

! active PPPoE sur l'interface.
! associe l'interface au **Dialer1**.

Configuration of Dialer1

- Cust1(config)# interface dialer 1
Cust1(config-if)# mtu 1492
Cust1(config-if)# ip address negotiated
Cust1(config-if)# encapsulation ppp
Cust1(config-if)# dialer pool 1
Cust1(config-if)# ppp authentication chap callin
Cust1(config-if)# ppp chap hostname Cust1
Cust1(config-if)# ppp chap password ciscoppoe
Cust1(config-if)# exit

! l'IP sera donnée par l'ISP via PPP.
! PPP est nécessaire pour PPPoE.
! correspond à dial-pool-number 1.

! doit être le même que dans ISP username.
! doit être identique au mot de passe ISP.

Default route :

- Cust1(config)# ip route 0.0.0.0 0.0.0.0 dialer 1

! tout le trafic Internet passe par **Dialer1**.

Configure debugging on the Cust1 router to display PPP and PPPoE negotiation.

Cust1# debug ppp negotiation

! PPP protocol negotiation debugging is on

Cust1# debug ppp authentication

! l'authentification **CHAP** et validation du username et password.

This displays:

- PPP negotiation
- CHAP authentication used by PPPoE.

Activation interface :

Cust1(config)# interface GigabitEthernet0/1
Cust1(config-if)# no shutdown

Configuration Complet de router ISP:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Cration de l'utilisateur pour CHAP
Router(config)#username Cust1 password ciscoppoe
Router(config)#
Router(config)#! Pool d'adresses IP pour les clients PPPoE
Router(config)#ip local pool PPPoE_POOL 10.0.0.1 10.0.0.10
Router(config)#
Router(config)#! Interface qui servira de template pour les clients
Router(config)#interface Virtual-Template1

Router(config-if)#ip unnumbered Gig0/1
Router(config-if)#mtu 1492
Router(config-if)#peer default ip address pool PPPoE_POOL
Router(config-if)#ppp authentication chap
AAA: Warning, authentication list default is not defined for PPP.

Router(config-if)#exit
Router(config)#
Router(config)#! Cration du groupe PPPoE
Router(config)#bba-group pppoe GLOBAL
Router(config-bba)#virtual-template 1
Router(config-bba)#exit
Router(config)#
Router(config)#! Activation PPPoE sur l'interface connecte au client
Router(config)#interface Gig0/1
Router(config-if)#pppoe enable group GLOBAL
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#write memory
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

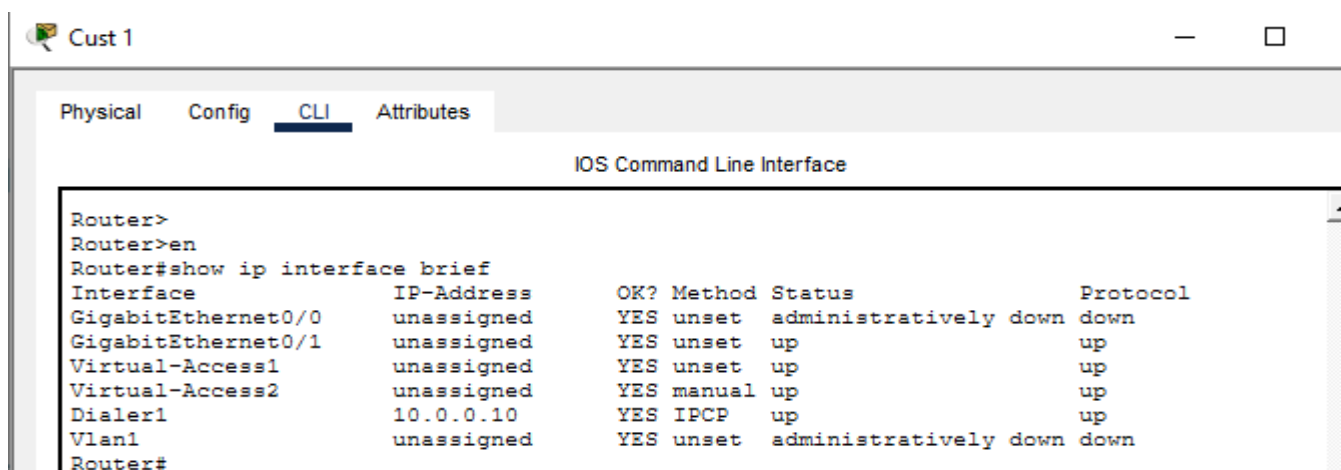
Configuration Complet de router Cust 1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Gig0/1
Router(config-if)#pppoe enable
Router(config-if)#pppoe-client dial-pool-number 1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Dialer1
Router(config-if)#mtu 1492
Router(config-if)#ip address negotiated
Router(config-if)#encapsulation ppp
Router(config-if)#dialer pool 1
Router(config-if)#ppp authentication chap callin
AAA: Warning, authentication list callin is not defined for PPP.
Router(config-if)#ppp chap hostname Cust1
Router(config-if)#ppp chap password ciscoppoe
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 Dialer1
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#
Router(config)#end
Router#write memory
Building configuration...
[OK]
Router#debug ppp negotiation
PPP protocol negotiation debugging is on
Router#debug authentication
%LINK-5-CHANGED: Interface Virtual-Access1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Virtual-Access1, changed state to up
```

Execute a (show ip interface) brief command on the Cust1 router to display the IP address assigned by the ISP router. By what method was the IP address obtained :

Answer: L'adresse IP est obtenue par négociation PPP (IPCP), donc la méthode est negotiated via PPPoE.

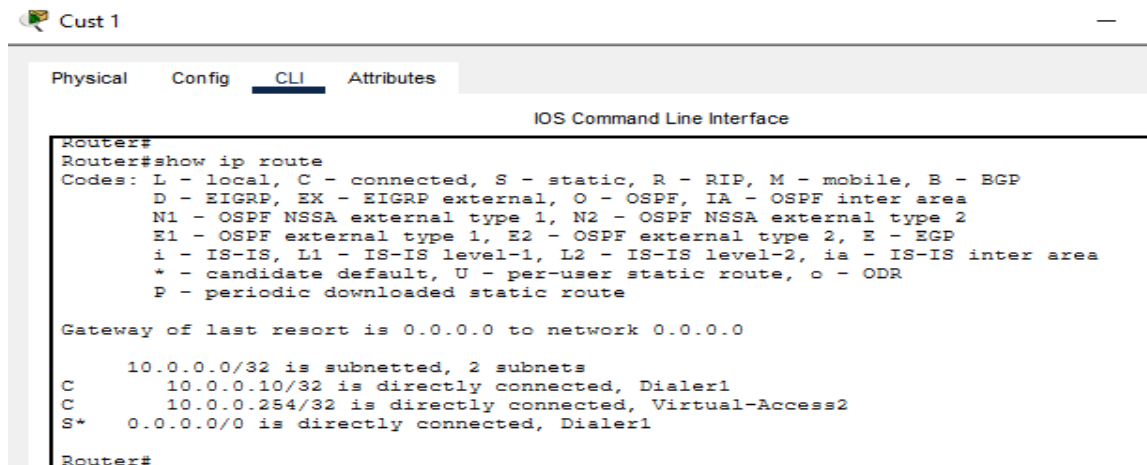


The screenshot shows a window titled 'Cust 1' with a tabbed interface. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The output of the 'show ip interface brief' command is as follows:

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up
Virtual-Access1	unassigned	YES	unset	up	up
Virtual-Access2	unassigned	YES	manual	up	up
Dialer1	10.0.0.10	YES	IPCP	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Execute a (show ip route) command on the Cust1 router :

Answer: The routing table displays the default route pointing to Dialer1, which is used to reach the ISP network.



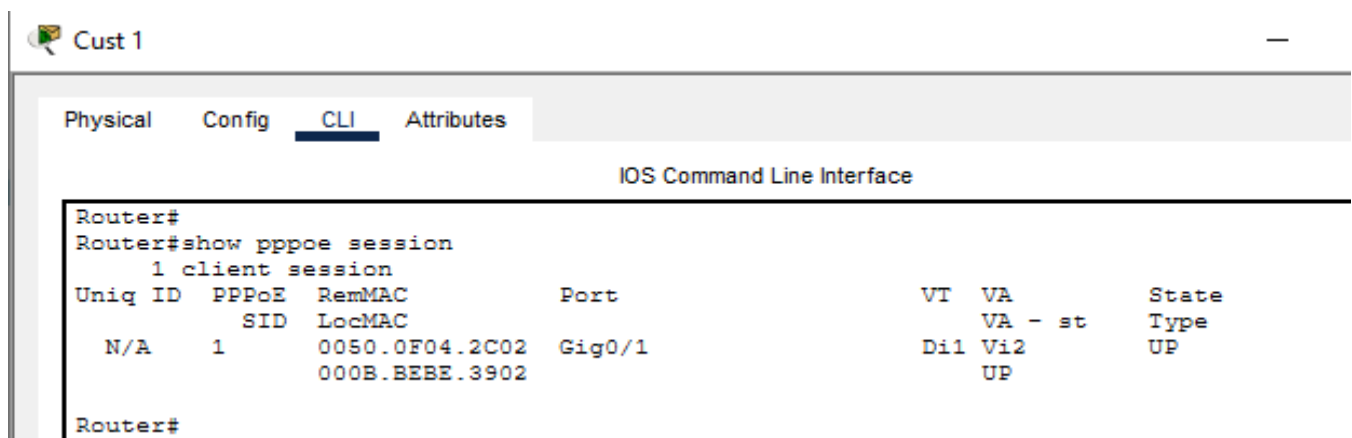
```
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    10.0.0.0/32 is subnetted, 2 subnets
C      10.0.0.10/32 is directly connected, Dialer1
C      10.0.0.254/32 is directly connected, Virtual-Access2
S*    0.0.0.0/0 is directly connected, Dialer1
Router#
```

Execute a (show pppoe session) command on the Cust1 router :

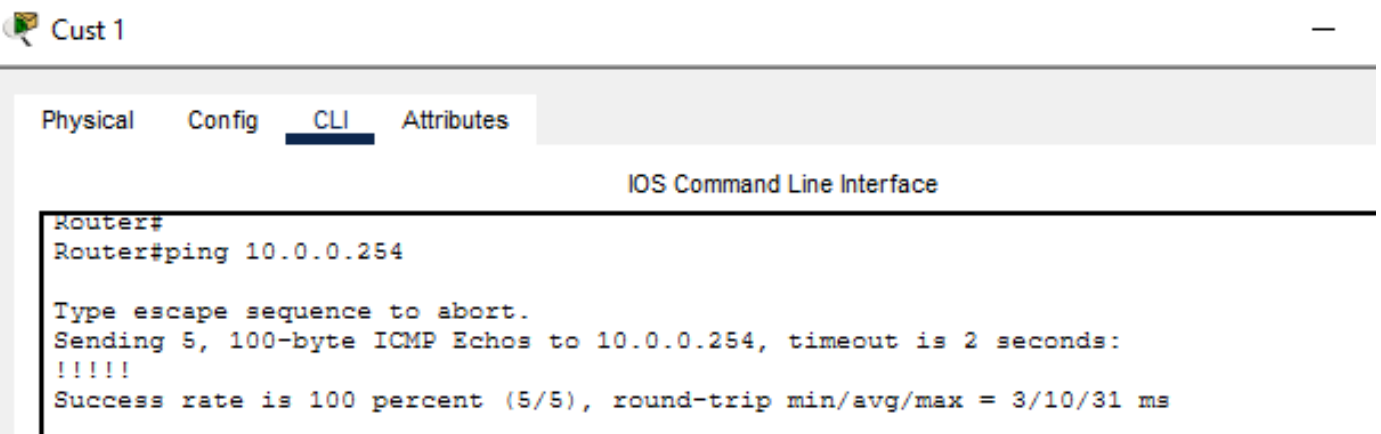
Answer : This command displays the active PPPoE session between the Cust1 router and the ISP router, confirming that the PPPoE connection is established.



```
Router#
Router#show pppoe session
1 client session
Uniq ID  PPPoE  RemMAC          Port          VT  VA  VA - st  State
      N/A    1          0050.0F04.2C02  Gig0/1        Di1 Vi2    UP
                   000B.BE8E.3902
```

Execute a ping request to 10.0.0.254 from the Cust1 router :

Answer : The ping should succeed, confirming that the PPPoE connection between Cust1 and the ISP is working correctly.



```
Router#
Router#ping 10.0.0.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.254, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/10/31 ms
```

Question Why do ISPs using DSL technology also primarily use the PPPoE protocol with their clients?

Answer :

- PPPoE allows **authentication of users** using protocols such as CHAP or PAP.
- It enables **dynamic assignment of IP addresses** to clients.
- It allows **account management and billing control** by the ISP.
- It works over **Ethernet networks used in DSL infrastructure**.

Conclusion

In this lab, we successfully configured a PPPoE connection between a client router and an ISP router. This experiment demonstrated how PPPoE is used in DSL networks to provide authentication and dynamic IP address assignment. The connectivity tests confirmed that the network configuration was correctly implemented.